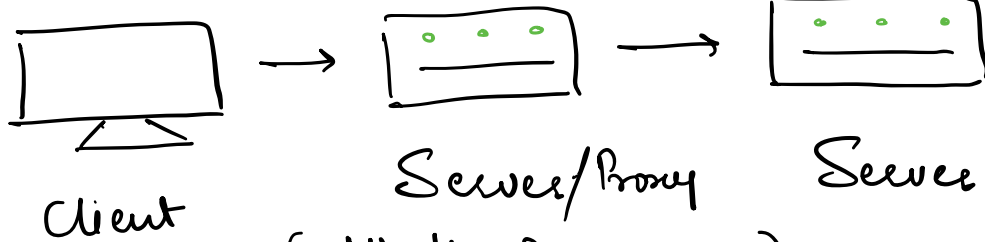


Proxies & Load Balancers

proxy = Forward proxy vs

Reverse proxy



(Middle Server
takes our request
& forwards it actual
Server)

Proxy:- Is kind of a middle man that does the work of the client and helps hide the client's identity

* Proxy Server

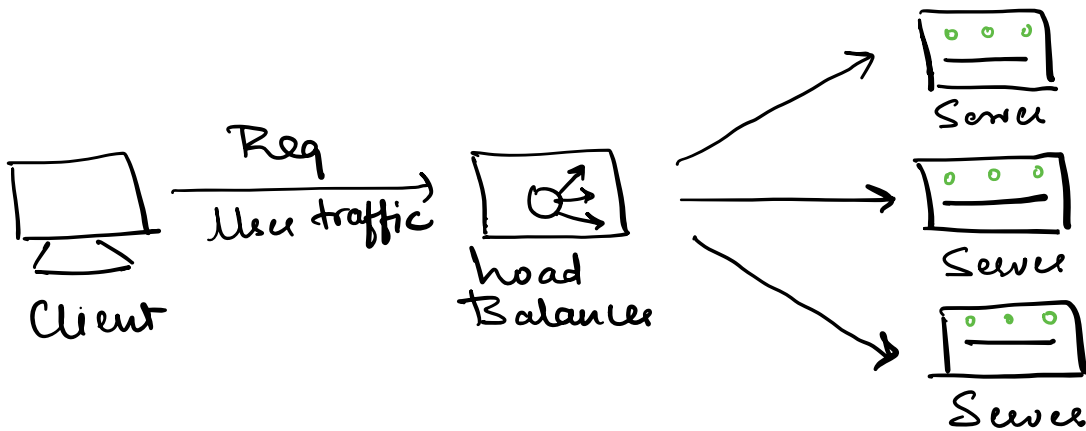
In good way, say you are not able to access the origin server but if you send req to proxy it does that work for you to access the origin server

Forward proxy → In bad way, say in corporate network proxy will block access to certain server (eg youtube.com) as client system have to be connected to a proxy eg: VPN - hide your IP address from the server

Reverse proxy → The proxy server takes the req from the client forwards it to the server the server fulfills the req and gives the response to proxy & proxy returns it back to the client. The client does not know the existence of the server

but the server might know about the client
eg: 1) CDN - we only interact with CDN server and
it will do the work for us
2) Load Balancer

Load Balancer



(As single cannot handle all the traffic)

Load Balancer Algorithms

1) Round Robin - The 1st req go to 1st server
2nd " " " 2nd "
3rd " " " 3rd "
4th " " " 4th "
⋮

Weighted Round Robin - 1st server weight 10 (2)
2nd " " 5 (1)
3rd " " 5 (1)

So the first server should get 2 times good
as the other two so should get double
requests

2) LB best connections -

Some req may take longer to process like the 1st server takes more time to complete a req but 2nd server finished a req (not occupied) so the latest req will be taken up by second server

3) LB User location

Route the req based on the nearest server eg user is in Asia than the server in Asia will be allocated

4) LB Hashing

Take a req use some fields like the IP address/content and take hash of it

Types of Load Balancers

Layer 4	vs	Layer 7
Lower network layer TCP		Application layer HTTP
- faster - look at IP only to balance eg Round Robin location - less flexible coz Not smart as don't have access to application data like not able to see the resource user is using		- slower - Intelligent routing Smart can look at the application data eg:- one server handle tweets, other server user data, third handles auth so route based on content - Multiple connection since need to decrypt the request

X. Multiple Replica of Load Balancer in case of crash or single point failure so horizontal scaling

Nginx - Powerful, opensource, implement our own

AWS, GCP - Cloud Balancer already available